# XT FIRMWARE FOR LXR-02

VERSION 1.2.13 · SUPPLEMENT TO THE LXR OWNERS MANUAL

### SAMPLES & WAVES

31 WAV samples, 8 single-cycle waves and 4 wavetables from the SD card, with start, length and loops.

### SEQUENCER

A step grid per track, ratchets and roll locks, trigger tracks, LFO start, lock trigger and instant pattern switching.

### EFFECTS

A second FX slot, sends, pre-fader option and a master bus on St1: saturation, compressor and EQ.

### ENGINES & MORE

Modal, FM2 and wavetable (test), kit and voice morph, cue, WAV recorder, INIT templates.

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github.com/shellbeat-jpg/lxr-desktop-sample
November 2026

# TABLE OF CONTENTS


3 **1 NOTES**
3 **1.1.** About the tables
3 **1.2.** What is stored where
3 **1.3.** Installation
3 **1.4.** Compatibility

4 **2 MEMORY MANAGEMENT**
4 **2.1.** SD card
4 **2.2.** Sample import
5 **2.3.** Experimental: 1.5 MB sample memory
5 **2.4.** Kits
5 **2.5.** INIT templates for voices

6 **3 VOICES**
6 **3.1.** Engines per voice
6 **3.2.** Oscillator sources
6 **3.3.** Oscillator page (OSC)
6 **3.4.** Samples and loops
7 **3.5.** Waves
7 **3.6.** Modulated sample selection
7 **3.7.** Amplitude envelope (AEG)
8 **3.8.** LFO page (LFO)
8 **3.9.** Track 7
8 **3.10.** LFO 2 (test)
8 **3.11.** Random voice
9 **3.12.** Engine modal (test)
9 **3.13.** Engine fm2 (test)
10 **3.14.** Engine wavetable (test)

11 **4 SEQUENCER**
11 **4.1.** Lock trigger: velocity 0
11 **4.2.** Step flam: ratchets and roll locks
11 **4.3.** Step accent and octave
11 **4.4.** Instant pattern switch
11 **4.5.** Step grid per track
12 **4.6.** Trigger tracks
12 **4.7.** LFO start
12 **4.8.** Start, loop, gates, copying (test)

13 **5 PERFORMANCE MODE**
13 **5.1.** Performance menu
13 **5.2.** WAV recorder
14 **5.3.** Kit morph
14 **5.4.** Voice morph
14 **5.5.** Cue on St2

15 **6 EFFECTS**
15 **6.1.** Signal flow
16 **6.2.** FX1 and FX2
16 **6.3.** Sends and pre-fader
16 **6.4.** FX knobs
17 **6.5.** Master bus FX3

18 **7 GLOBAL SETTINGS**

18 **8 MIDI**

19 **A WHERE SETTINGS ARE STORED**

20 **B MODULATION TARGETS**

# 1 > NOTES

This supplement describes what the **XT firmware 1.2.13** (formerly OE02-Samples+) adds to the LXR-02 and its manual.

It contains the modifications of **OpenElixir-02 1.2** (Moritz Waser), marked OE.

## 1.1. ABOUT THE TABLES

As in the manual: **display name** (three letters), full name, function. The edit view (press the encoder) shows the full name and the value.

Buttons by their label below the step button, e.g. **SHIFT + FX** (step 15).

## 1.2. WHAT IS STORED WHERE

**Global**: kept only after SAVE → Global. All new settings with their storage: appendix A.

## 1.3. INSTALLATION

1. **LXR_XT_V1.2.13.img** into the top folder of the card; delete all other **LXR*.img** there.
2. Switch on with the encoder button held until "LXR Updater" appears.
3. "Update OK!", then delete the image.
4. The update erases the samples: leave **SAMPLES** on the card, the next start imports them again.

After switching off, wait ≈ 10 s.

## 1.4. COMPATIBILITY

Kits and patterns of the original firmware and of OpenElixir-02 load; new settings start with default values.

New kits also load in older firmwares (without the additions); voices with a sample or wave stay silent there, new engines play the default engine.

Going back: install the image of OpenElixir-02 or Erica Synths the same way.

# 2 > MEMORY MANAGEMENT

## 2.1. SD CARD

Besides the projects (and **REC**, chapter 5.2):

| Folder / file | Contents |
|---|---|
| SAMPLES | one-shot samples, WAV |
| SAMPLES/LOOPS | loop samples, WAV |
| SAMPLES/WAVES | up to 8 single-cycle waves, WAV |
| SAMPLES/TABLES | up to 4 wavetables, WAV (3.14) |
| BIGFLASH | in **SAMPLES**: 1.5 MB memory (2.3) |
| INIT.SND | INIT template: top folder (global), project folder (2.5) |

## 2.2. SAMPLE IMPORT

At start the LXR compares the folders with the internal flash (names and sizes) and imports again on changes. The samples also play without a card.

| Rule | |
|---|---|
| Format | WAV, 16-bit PCM, mono or stereo (mixed to mono), any sample rate |
| Names | 8.3 format (**01KICK.WAV**); the slots **S1** … **S31** follow the order of the names, numbers by value (**2KICK** before **10KICK**) |
| Count | up to 31 samples (both folders together) and 8 waves |
| Length | about 5.9 s mono in total at 44.1 kHz (11.9 s at 22.05 kHz); a stereo file counts at half its size |
| Ignored | names starting with **.** or **_** |

A file replaced by one with the same name and size is not imported again.

### START SCREEN

**S:12 W:3 T:2** = stored samples, waves and wavetables.

Below: **Used 4.2/5.9s** = used and available sample time.

## 2.3. EXPERIMENTAL: 1.5 MB SAMPLE MEMORY

The chip reports 1 MB of flash but usually has more.
With **BIGFLASH** in the **SAMPLES** folder: **17.8 s** instead of 5.9 s.

- Not guaranteed (ST). Every sample is verified; on an error "Big flash failed / using 512K" and import into the normal memory.
- If the start hangs permanently: remove **BIGFLASH** or start without a card. Firmware and bootloader are not affected.

## 2.4. KITS

Additionally in the kit: per voice the engine OE, start and length of the sample, choke group, sends; per kit the order of the FX slots and FX3.

## 2.5. INIT TEMPLATES FOR VOICES

Hold **LOAD** + step button of the voice (1–7), one step back from kit 0: **INIT: Project** / **Global** / **Factory** shows the source. Click:

| Action | Description |
|---|---|
| Load INIT | from the project template, else the global one, else the engine defaults |
| Save to Project | voice into the project template |
| Save to Global | voice into the global template |
| Del Project INIT, Del Global INIT | remove the voice from the template (click again) |

Each complete voice (engine, sound, LFO, velocity, sample). Project template: only in saved projects.

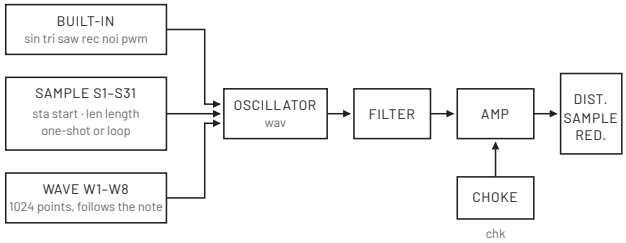
# 3 > VOICES

## 3.1. ENGINES PER VOICE OE

Each of the 6 slots selects its engine: drum, snare, cymbal, hi-hat.
New: **modal**, **fm2** and **wavetable** (3.12–3.14).
**SHIFT + KIT**: engine menu of the active voice. In the kit.

## 3.2. OSCILLATOR SOURCES

Main oscillator: built-in waveforms, **samples** (own pitch, once or as a loop) or **waves** (single cycles, follow the note).




## 3.3. OSCILLATOR PAGE (OSC)

| Display | Name | Description |
|---|---|---|
| wav | Waveform | **sin** ... **pwm** (standard)<br>**S1** ... **S31** (samples)<br>**W1** ... **W8** (waves)<br>Edit view: file name, **L** = loop, slot |
| sta* | Start | ms, 0 = beginning (all voices, replaces **pwm**) |
| len* | Length | ms, 0 = to the end (drum 1–3) |
| coa | Coarse tune | with a sample: 60 = original pitch |
| oct | Octave rnd | random octave per trigger: **±1**, **+1**, **–1** ... **±3**, **+3**, **–3** (0 included), **off**; step value: 4.3 |

*: samples only; taken at the trigger, also with LFO/velocity.

FM page of the drum engine: samples and waves also as FM source.

## 3.4. SAMPLES AND LOOPS

- **SAMPLES**: once per trigger; the AEG decay can end it earlier.
- **SAMPLES/LOOPS**: looped from **sta** to **len**. Each trigger restarts at **sta**; in between the loop runs until the decay fades it out (**dec** = 127: until the next trigger).
- Sequencer stop fades out all loops.

## 3.5. WAVES

- Up to 8 WAV files in **SAMPLES/WAVES**, one cycle each: **W1** … **W8** (W1 = first name). Stereo: left channel.
- Against clicks: start and end at a zero crossing, rising at both ends, no DC offset.

## 3.6. MODULATED SAMPLE SELECTION

An LFO or velocity with target **wav** selects among the following samples (or waves):

- Range: from the selected sample, as many more as LFO **mod** / velocity **amt** (S5, amount 4 = S5 … S9). Chosen at the trigger.
- **Velocity layers**: soft to loud versions, named in sequence, select the softest, **amt** = layers – 1.
- **Round robin**: LFO **dst** = **wav**, LFO waveform **cyc**: one sample per trigger.
- **Random**: LFO waveform **rnd**, rate faster than the triggers.
- Only the target **wav** selects samples, not **fwv** / **cwv**.

## 3.7. AMPLITUDE ENVELOPE (AEG)

| Display | Name | Description |
|---|---|---|
| chk | Choke | choke group of the track: **off**, **A**, **B**, **C**; edit view: the voice buttons move the tracks to the next group, e.g. **AA-B-CC** |
| sna | FX1 send | send into slot A, 0 … 127 |
| snb | FX2 send | send into slot B, 0 … 127 |

- **Choke**: a trigger cuts the other tracks of its group. Tracks 6 and 7 cut each other anyway.
- **Sends**: after pan, after the fader (unless **pre** is on, chapter 6.3). Send/return: slot at 100 % wet.

## 3.8. LFO PAGE (LFO)

| | |
|---|---|
| wav | New waveforms |
| cyc | step counter: one step per trigger,<br>**mod** = count (round robin) |
| xtr | exponential triangle |
| si1 tr1 su1 sd1 sq1 | one-shot: sine, triangle, saw up/down, square |
| xu1 xd1 xt1 | one-shot exponential: up, down, triangle |
| rn1 | new random value per trigger |
| w1 … w8 / w1. … w8. | waves (running / one-shot) |

| Display | Name | Description |
|---|---|---|
| ofs | Offset | one-shot: delay until the start<br>(64 ≈ half a cycle) |
| mod | Amount | edit view: range of the target, e.g. **21..127** |
| dst | Destination | also rate, amount … of **any** other LFO |

One-shots: one cycle per trigger (voice of **rtg**, else its own), then hold. Length: **frq** / **snc**.

## 3.9. TRACK 7

Tracks 6 and 7 play one voice. Track 7 has its own values for decay, **wav**, **coa** (±24 semitones to track 6), **sna**, **snb**, **pan** and its own LFO. Edit view: "Trk7 …", **=t6** = as track 6. Sends and pan: those of the track that triggered last.

## 3.10. LFO 2 (TEST)

- Second LFO screen, press LFO again: **LFO 2** of the voice, same parameters.
- Track 7: the LFO page is its own LFO; acts while track 7 plays.
- Stored in the kit, not morphed.

## 3.11. RANDOM VOICE

- **SHIFT + encoder click** on a voice page (OSC, AEG, MOD, FM, transient, filter; not in the edit view): new random variant of the active voice.
- Amount: **rnd** (PERF, default 32). Switching values change with this probability.
- Kept: volume, pan, **out**, sends, LFO, choke, engine.
- Undo: reload the voice.

## 3.12. ENGINE MODAL (TEST)

Physical modelling: 6 resonators ring in the modes of a struck body. Drum pages:

| Display | Name | Description |
|---|---|---|
| dmp | Damping | MOD: decay time, 30 ms … 6 s |
| inh | Inharmonic | MOD: stiffness, spreads the upper modes |
| pos | Strike pos. | MOD: edge (0) … centre (127) |
| hrd | Hardness | FM: soft … hard mallet |
| mat | Material | FM: 0 = metal … 127 = wood (highs decay faster) |
| mdl | Model | FM: **mem** membrane, **bar** bar, **str** string, **bel** bell, **plt** plate, **gls** glass |
| exc | Exciter | FM: **mal** mallet, **nse** noise |

The transient sample excites in addition. A new strike adds to the ringing. New voice: decay 110.

## 3.13. ENGINE FM2 (TEST)

2-operator FM: sine carrier, sine modulator with its own index decay. FM page:

| Display | Name | Description |
|---|---|---|
| rat | Ratio | modulator : carrier, **.50** … **11** (odd values = metallic) |
| idx | FM index | modulation depth at the strike |
| idc | Index decay | 5 ms … 4 s |
| fbk | Feedback | modulator onto itself, towards noise |

MOD page: pitch EG as on the drum.

## 3.14. ENGINE WAVETABLE (TEST)

Scans a set of waves (frames). FM page:

| Display | Name | Description |
|---|---|---|
| set | Table set | **bas** sine › triangle › square › saw, **usr** from the waves, **all** = both, **t1** … **t4** from **SAMPLES/TABLES**. Edit view: file name or **none>bas** |
| pos | Wave pos. | 0 = first frame … 127 = last |
| swp | Pos. sweep | the pitch EG pushes the position up at the strike, back to **pos** |
| scn | Scan mode | **smo** smooth, **stp** stepped, **pp** back and forth, **pps** back and forth stepped |

- **Position** = **pos** + **swp** × pitch EG (MOD page: decay, slope; amount 0 = no pitch change) + LFO/automation on **pos**. Not the amp envelope.

OWN SETS: SAMPLES/TABLES

- Up to 4 WAV files (**t1** = first name): 16/24 bit or float, mono/stereo (left).
- Cycle length: from the **clm** chunk (Serum, Vital), else 2048 or 256 (WaveEdit).
- Max. 64 frames; fewer: in-between waves are added. 64 frames ≈ 0.9 s of sample memory, after the samples.

# **4** > SEQUENCER

## 4.1. LOCK TRIGGER: VELOCITY 0

Velocity 0: no trigger, no MIDI note; **P1**/**P2** still act ("trigless lock"). SHIFT + encoder click on the automation page clears both locks.

## 4.2. STEP FLAM: RATCHETS AND ROLL LOCKS

| flm | Effect (after x2 … x8) |
|---|---|
| x12, x24 | 12 / 24 repeats in the step |
| x4+, x4- | 4 repeats, velocity rising / falling (25 … 100 %) |
| 1/8 1/4 1/2 1/1 | roll lock: a repeat every 2, 4, 8, 16 steps up to the next active step |

## 4.3. STEP ACCENT AND OCTAVE

- **acc** (first step page, after **prb**): accent on/off for the step, stored in the pattern.
- **aca** (CONFIG, second page, global): amount +0.1 … +6 dB, default 64 ≈ +3.5 dB.
- Acts like the voice's fader, only for this trigger.
- **oct** (after **sft**): random octave instead of the voice's (3.3): **vce** (as the voice), off, ±1, ±2, ±3, +1, −1.

## 4.4. INSTANT PATTERN SWITCH

**ips** on (CONFIG, second page): a pattern chosen by hand starts at the next step at the same position. Chains and songs still switch at the end.

## 4.5. STEP GRID PER TRACK

SHIFT + LAST STEP: **[16] 1/16 1** = length, grid, step length (1/16 = 1). Press the encoder: edit the grid. ALL mode (LAST STEP again): all tracks.

| Grid | Step length · steps per bar |
|---|---|
| 1/4 · 1/4T · 1/8 | 4 · 8/3 · 2 — 4 · 6 · 8 |
| 1/16. · 1/8T · 5/64 | 6/4 · 4/3 · 5/4 — 10⅔ · 12 · 12⅘ |
| 1/16 | 1 — 16 (default) |
| 1/32. · 1/16T | 3/4 · 2/3 — 21⅓ · 24 |
| 1/32 · 1/32T | 1/2 · 1/3 — 32 · 48 |

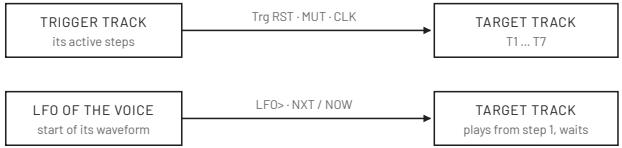
- Each track loops in its length (polymeter); the pattern ends after the longest.
- **rol** and flams count in the track's steps; live recording in 1/16.
- Stored in the pattern.

## 4.6. TRIGGER TRACKS

LAST STEP page, each encoder click moves on: length › **Start** › grid › **Trg [OFF]** › target **>[T3]** › **LFO>[T3]** › **NXT**/**NOW**. Taken targets are skipped.

| Trg | Each active step … |
|---|---|
| RST | sets the target to step 1 (plays at once) |
| MUT | mutes the target for the step's duration |
| CLK | advances the target one step (only with these pulses) |

All velocities count; velocity 0 does not sound itself. A track is the target of one source at most.




## 4.7. LFO START

**LFO>[T3]**: each start of the LFO waveform of this track's voice (new cycle, **rtg**, one-shot) starts track 3 at step 1: **NXT** at the next step in the target's grid, **NOW** at once.

The target plays to its last step and waits (silent until the first start). With **snc** in time, free it drifts. Trigger and LFO target at once.

## 4.8. START, LOOP, GATES, COPYING (TEST)

- **Start** (LAST STEP page): track plays from there, also after RST/LFO.
- **Loop** per track: **SHIFT + PROJECT** on/off for the active track. Hold a step, press a second (also another page); second before first = backwards. Single press or pattern change: off.
- **Gate**: hold a step, press a later one: the first holds to the end of the second (AEG full, note off at the end).
- **Prepared**: hold the voice button + step to mark; while off it flashes on the beat.
- **Copy**: hold **COPY**, select steps, release; each step press pastes them there, also into tracks of the same engine. **COPY** ends.
- **SHIFT + MORPH**: duplicate the range start … last step behind it (× 2, max. 64).*
- **COPY + ‹ / ›**: undo / redo the last pattern change (one step).*

*: Deviation: in the original, **COPY + ›** fills the next bar.

# **5** > PERFORMANCE MODE

## 5.1. PERFORMANCE MENU

| Display | Name | Description |
|---|---|---|
| rol | Roll | new **24**, **32**, **48** (1/16T, 1/32, 1/32T); in the track's steps |
| dj res | DJ filter OE | LP/HP and resonance on the final mix |
| sat frq | Saturation, split | from FX3 (chapter 6.5) |
| vmv vmo vsw | Voice morph | chapter 5.4 |
| osx | OSR exclude | **on**: voice ignores the global **srt**/**0sr**; edit view: voice buttons |
| cmx cvl | Cue mix, level | chapter 5.5 |
| rnd | Random | amount of the random voice (3.11) |
| rec rst | Record WAV | page 3, chapter 5.2 |

## 5.2. WAV RECORDER

Records St1 (after FX3, DJ filter, limiter) as WAV to the card: **REC/REC001.WAV** ..., 16-bit stereo, 44.1 kHz.

| Display | Name | Description |
|---|---|---|
| rec | Record WAV | length: **x1**, **x2**, **x4** passes of the longest track, **end** until stop. Display **arm** / **rec** |
| rst | Rec start | **ply** with PLAY, **pat** at the next pattern start (also with the sequencer running) |

- **Encoder click** in the edit view: arm (creates the file) / cancel / end the recording.
- Ends after the passes, at STOP or on a click: "Saved REC003 1:32".
- **rec** and **rst** are global and stay set after the recording.
- Slow card: dropouts are counted ("2 gaps"). Avoid loading/saving while recording.

## 5.3. KIT MORPH

- MIDI CC 1 (global channel) sets **mrp**.
- **mrp** 0 → 127: from the active kit to the morph kit. Edit view + **SHIFT**: **Morph>** and target value; **SHIFT + encoder click**: parameter does not morph.
- **mrv** (CONFIG, page 2): morph per voice; edit view: voice buttons.

## 5.4. VOICE MORPH

| Display | Name | Description |
|---|---|---|
| vmv | Voices | pair source > target: OFF, 1>2 … 6>5 |
| vmo | Amount | 0 = source, 127 = values of the target |
| vsw | Switch | from this **vmo** on: switching values (waveform, filter type) of the target |

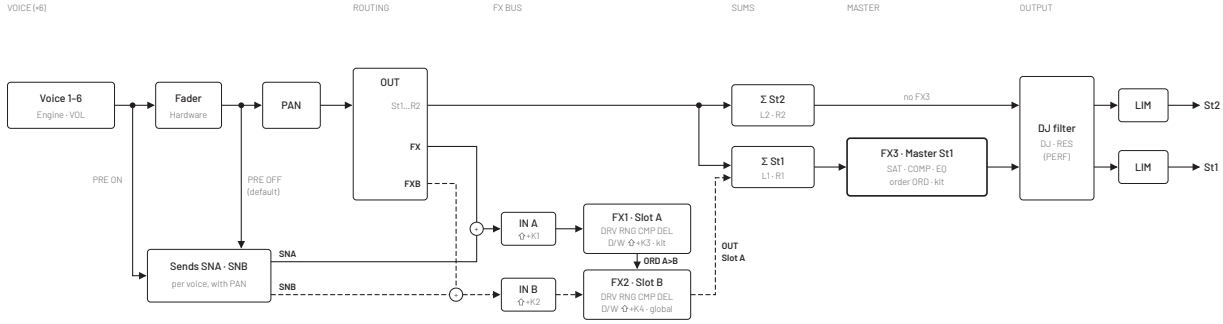
Only **vsw** is stored (global).

## 5.5. CUE ON ST2

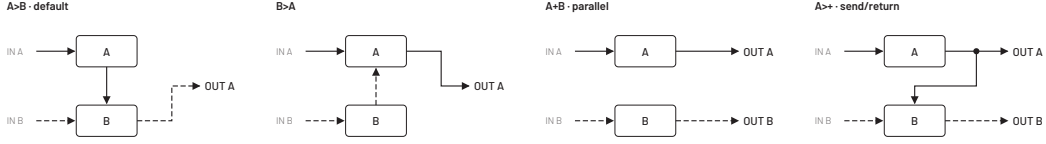
- **SHIFT + step 1–7** (on any page): pre-listen to a track, LED blinks.
- Pre-fader, also muted tracks. St2 becomes the cue bus automatically.
- **cmx**: cue ↔ main on St2, **cvl**: level.
- Needs cables in St1 (L and R); otherwise St1 plays on the St2 jacks.

# 6 > EFFECTS

## 6.1. SIGNAL FLOW




From the voice to the jacks, slot order A>B (dashed: slot B; ⇧ = SHIFT, K1 … K4 = parameter knobs). The fader acts on the sends only with **pre** off. FX3 sits on St1 (without a cable in St1 on the St2 jacks).




Order of the FX slots, **ord** on the FX1 page (chapter 6.2).

## 6.2. FX1 AND FX2

**SHIFT + FX**: FX1 (kit, slot A) › FX2 (project, slot B) › FX3 (master) › FX1. Both slots: **off drv rng cmp del**.

| Display | Name | Description |
|---|---|---|
| typ / fx2 | Type | effect of slot A / B |
| out | Audio out | output of the slot |
| d/w | Dry/Wet | mix; also SHIFT + knob 3 / 4 |

Setup page (turn past the last parameter):

| Display | Name | Description |
|---|---|---|
| sya / syb | Sync A / B | delay time as a note value, 1/4 … 1/32T |
| ord | Order | FX1 page: routing of the slots |
| prs | Preset | click, choose, click: 17 factory presets (delay, drive, compressor, ring), **U1**–**U8** per project, **Save to U**n |

| ord | Routing |
|---|---|
| A>B | A into B, to the output of A (default). B is an insert: at 100 % wet the dry sound of A is missing. |
| B>A | B into A, to the output of A |
| A+B | parallel, each to its own output |
| A>+ | send/return: A to its output and into B, B to its output |

FX1 and **ord**: kit. FX2 and sync: global.

## 6.3. SENDS AND PRE-FADER

**sna** / **snb** (AEG page): sends into slot A / B. MIX **out** = **FXB**: 100 % into slot B.

| Display | Name | Description |
|---|---|---|
| pre | Pre-fader sends | CONFIG, page 2. **off**: sends after the fader. **on**: before it, the fader sets only the dry signal (fader down = only the echo). |

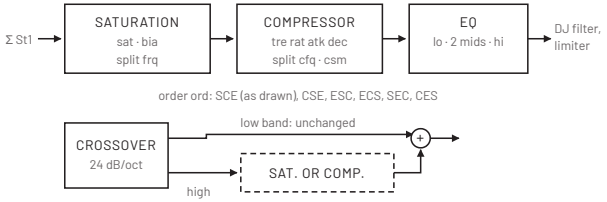
**vol** (MIX) always acts on both.

## 6.4. FX KNOBS

**SHIFT** + knob 1 / 2: input level FX1 / FX2 (−∞ … +12 dB, global); 3 / 4: mix. Not on LOAD/SAVE.

## 6.5. MASTER BUS FX3

Third slot at the end of **St1**: after FX1/FX2, before DJ filter and limiter. Saturation (S), compressor (C), EQ (E) in the order **ord**. In the kit.




Split: only the band above the crossover is saturated / compressed. **csm** = **sc**: the upper band only controls, everything is compressed.

FX3, PAGE 1

| Display | Name | Description |
|---|---|---|
| sat | Saturation | 0 = off; soft clipping up to +24 dB, level compensated |
| frq | Sat split | off or 40 Hz … 10 kHz |
| cfq | Comp split | off or 40 Hz … 10 kHz |
| csm | CFQ mode | **spl** band split, **sc** sidechain filter |
| tre | Threshold | –48 … 0 dB |
| rat | Ratio | 0 = off, 1:1 … 20:1 |
| atk | Attack | 0.1 … 100 ms |
| dec | Release | **aut** = 1/16 of the tempo, else 20 … 1000 ms |

FX3, PAGE 2: EQ

| Display | Name | Description |
|---|---|---|
| lo | Low shelf | 100 Hz, ±15 dB (64 = 0 dB = off) |
| f1 g1 q1 | Mid 1 | 20 Hz … 16 kHz, ±15 dB, Q 0.3 … 8 (300 Hz, Q 1) |
| f2 g2 q2 | Mid 2 | as mid 1 (3 kHz) |
| hi | High shelf | 8 kHz, ±15 dB |

FX3, PAGE 3

| Display | Name | Description |
|---|---|---|
| ord | Order | SCE (default), CSE, ESC, ECS, SEC, CES |
| amu | Auto gain | **off** · **fix** half the reduction of a 0 dBFS peak · **ada** average reduction (≈ 1.5 s) |
| gai | Gain | make-up 0 … +24 dB, in addition to **amu** |
| d/w | Dry/Wet | compressor mix (parallel) |
| bia | Sat bias | 0 = symmetric; higher: even harmonics |

NOTES

- **sat**, **rat**, **gai** at 0 and EQ at 0 dB: FX3 off.
- Band split: **amu**, **gai**, **d/w** only on the compressed band.
- **amu** = **ada**: **d/w** changes only the character, not the loudness.
- **bia** only with **sat** > 0; for a mix 20 … 50 with **frq** 150 … 300 Hz.

EDIT VIEW

Top: peaks St1 / St2 before FX3 (dBFS). Bottom: **byp** = off, **on-4A** = 4 dB reduction, St1 (**B**: St2 jacks).

# **7** > GLOBAL SETTINGS

| Display | Name | SHIFT + CONFIG, page 2 |
|---|---|---|
| lim | Limiter OE | on every output |
| ips | Instant pattern | chapter 4.4 |
| mrv | Morph voices | kit morph per voice |
| pre | Pre-fader sends | chapter 6.3 |
| aca | Accent amount | chapter 4.3 |

# **8** > MIDI

| Message | Parameter (voice 1–6) |
|---|---|
| CC 1 · 2–5, 7, 8 | **mrp** (global channel) · OSC **wav** |
| CC 122–127 | LFO **mod** |
| NRPN 27–32 · 57–62 | LFO **wav** · LFO **ofs** |
| NRPN 87–92 · 100–105 | MIX **out** · OSC **sta**; **len**: 93, 94, 97 |
| NRPN 98 | engine OE: slot × 4 + engine, 24/30/36 + slot: modal/fm2/wt |
| NRPN 99 · 106–117 | **chk** · FX1 OE |

# **A** > WHERE SETTINGS ARE STORED

| Stored in | Settings |
|---|---|
| Kit | engine OE and its settings, **wav** (sample / wave number), **sta**, **len**, **chk**, **sna**, **snb**, **out**, LFO waveforms and **ofs**; FX1, **ord** of the FX slots; all of FX3; track 7, LFO 2, **osx**, **oct** |
| Pattern | step grid, flam values, accents **acc**, **oct** of the steps, start, trigger tracks, LFO start, steps with velocity 0 |
| Global | FX2, **sya**, **syb**, the input levels (SHIFT + knob 1 / 2), **pre**, **ips**, **mrv**, **vsw**, **cmx**, **cvl**, **aca**, **rnd**, **rec**, **rst**, **lim** OE |
| Project | the INIT template for voices, the values of the morph kit, FX presets **U1**–**U8** |
| Flash | samples, waves and wavetables (imported from the card) |
| Not stored | **vmv**, **vmo**, **dj**, **res** OE |

## MORE INFORMATION

The full description, with more examples and the history of each version, is in the repository:

- **README**: overview of all features and the installation.
- **docs/SAMPLES.md**: guide to all additions.
- **docs/OVERVIEW.md** / **OVERVIEW.de.md**: one-page overview in English and German.
- **docs/ENGINES.md**: the voice engines OE.
- **docs/MOD_TARGETS.md**: the modulation targets of each voice.

github.com/shellbeat-jpg/lxr-desktop-sample

The XT firmware is free software under the GNU General Public License 3.0. It is based on the LXR firmware by Julian Schmidt (Sonic Potions) and Erica Synths and contains the modifications of OpenElixir-02 1.2 (Moritz Waser). Font: Barlow, SIL Open Font License.

# **B** > MODULATION TARGETS

Targets of the LFOs, velocity and the step automation (parameter lock 1, 2). LFO and velocity **dst** show the name, the step page the voice and two letters (**1wa** = **wav** of voice 1, **0sr**: all voices). Targets the slot's engine lacks: **off**.

| Page | Name | Edit view | Voices |
|---|---|---|---|
| Oscillator | **coa** | Oscilltr Coarse | 1–6 |
| | **fin** | Oscilltr Fine | 1–6 |
| | **wav** | Oscilltr Waveform | 1–6 |
| | **pwm** | Oscilltr PulsWdth | 1–6 |
| | **noi** | Noise Frequncy | 4 |
| | **mix** | Nois/Osc Mix | 4 |
| Amp EG | **atk** | Veloc EG Attack | 1–6 |
| | **dec** | Veloc EG Decay | 1–5 |
| | **slp** | Veloc EG Slope | 1–6 |
| | **rpt** | Veloc EG RepeatCt | 4, 5 |
| | **d1** | Veloc EG Dcy Clsd | 6 |
| | **d2** | Veloc EG Dcy Open | 6 |
| Modulation | **dec** | Mod EG Decay | 1–4 |
| | **slp** | Veloc EG Slope | 1–4 |
| | **mod** | PitchMod Amount | 1–4 |
| | **dst** | Velocity DstParam | 1–6 |
| | **amt** | Velocity Amount | 1–6 |
| | **vol** | Velocity Vol mod | 1–6 |
| FM | **amt** | FM Amount | 1–3 |
| | **frq** | FM Frequncy | 1–3 |
| | **fwv** | Oscilltr Waveform | 1–3, 5, 6 |
| FM | **mod** | Oscilltr Mode | 1–3 |
| | **f1** | Mod Osc Freqcy 1 | 5, 6 |
| | **f2** | Mod Osc Freqcy 2 | 5, 6 |
| | **g1** | Mod Osc Gain 1 | 5, 6 |
| | **g2** | Mod Osc Gain 2 | 5, 6 |
| Transient | **cwv** | Transnt Waveform | 1–6 |
| | **vol** | Transnt Volume | 1–6 |
| | **frq** | Transnt Frequncy | 1–6 |
| Filter | **frq** | Filter Frequncy | 1–6 |
| | **res** | Filter Resnance | 1–6 |
| | **typ** | Filter Type | 1–6 |
| | **drv** | Filter Overdriv | 1–6 |
| LFO | **frq** | LFO Frequncy | 1–6 |
| | **snc** | LFO ClockSnc | 1–6 |
| | **mod** | LFO Amount | 1–6 |
| | **wav** | LFO Waveform | 1–6 |
| | **rtg** | LFO Retriggr | 1–6 |
| | **ofs** | LFO Offset | 1–6 |
| Mix | **srt** | Voice SampleRt | 0 (all), 1–6 |
| | **vol** | Voice Volume | 1–6 |
| | **pan** | Voice Panning | 1–6 |
| | **drv** | Voice Overdriv | 1–6 |
| FX | **p1** | Fx Param 1 | 1–6 |
| | **p2** | Fx Param 2 | 1–6 |
| | **p3** | Fx Param 3 | 1–6 |
| | **p4** | Fx Param 4 | 1–6 |

**fwv** = FM / mod oscillator, **cwv** = transient. A **wav** lock: 0–5 waveforms, 6–36 samples S1–S31, 37–44 waves W1–W8. All voices: docs/MOD_TARGETS.md.